DAA COLLEGE ASSIGNMENT

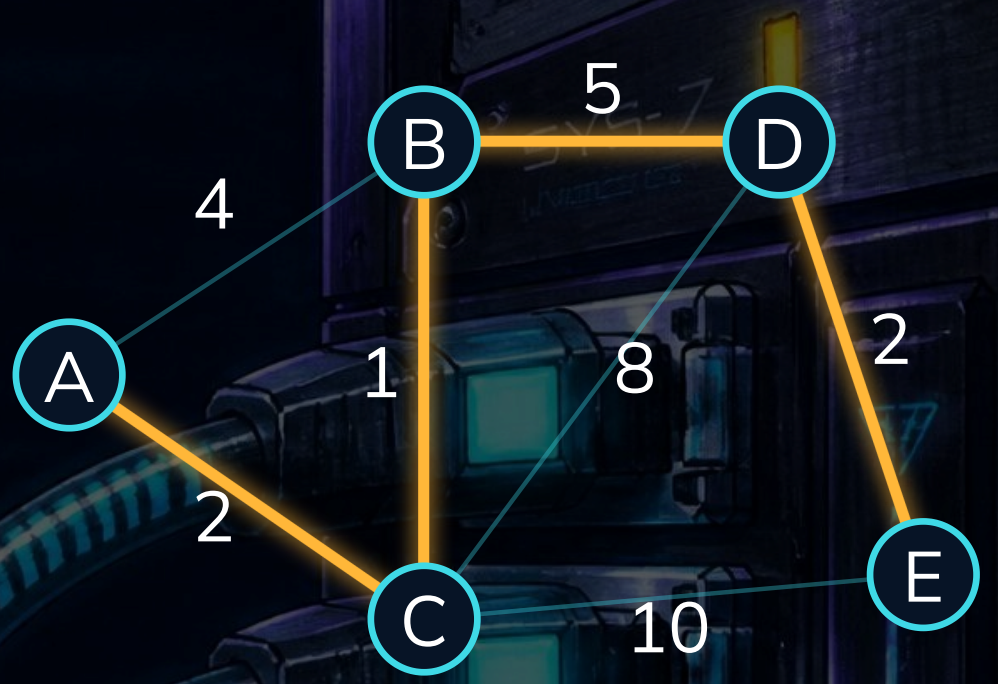
# Prim’s Minimum Spanning Tree Algorithm

## Building the Minimum-Cost Network

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Section D · 3rd Semester
Computer Science Engineering

Subject: Design and Analysis of Algorithms

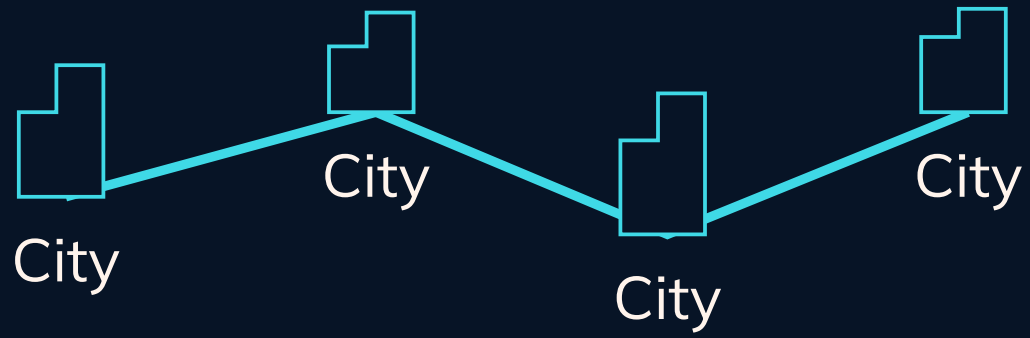



MINIMUM-COST NETWORK

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# What is a Minimum Spanning Tree?

A tree that connects all vertices of a connected weighted graph with the minimum possible total edge weight and no cycles.




ALL vertices

NO cycles

MINIMUM cost

For $V$ vertices, an MST has $V - 1$ edges.

Connecting cities with roads while spending the least possible money.




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# What is Prim’s Algorithm?

A greedy algorithm that finds the MST of a connected weighted graph.

01

### Start

Any vertex

02

### Find minimum edge

Visited → unvisited

03

### Add vertex

Cheapest valid edge

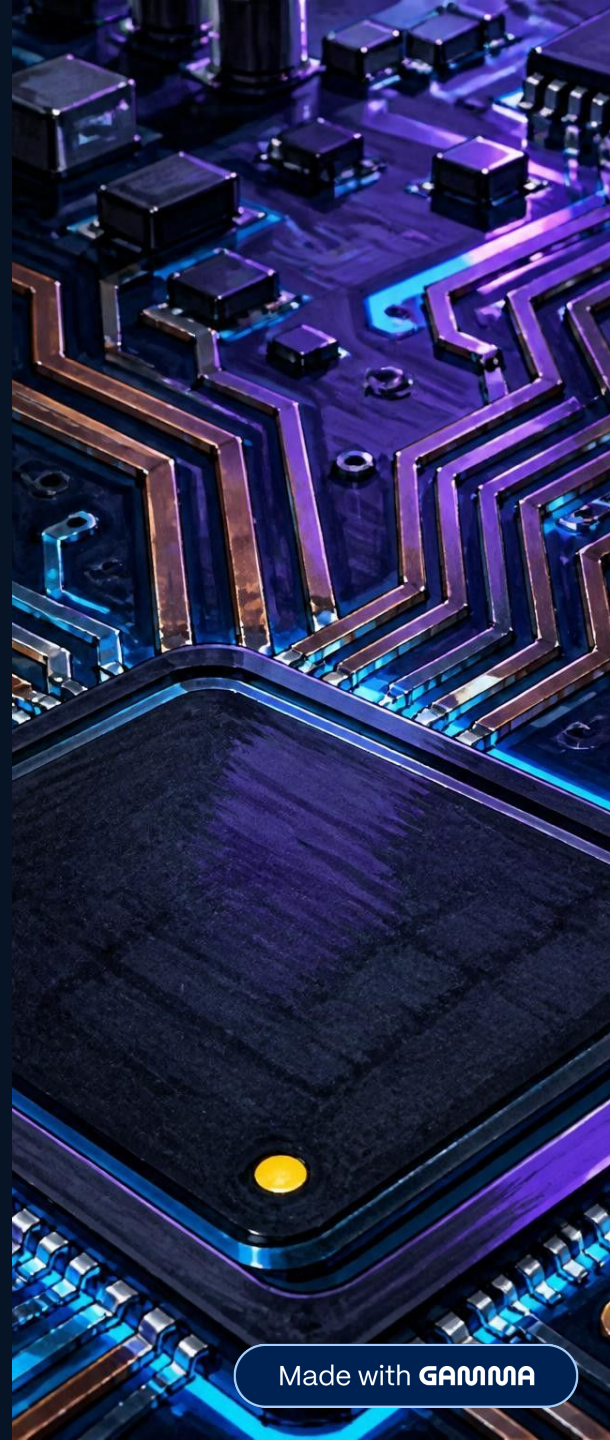
04

### Repeat

Build step by step

### A new vertex every time

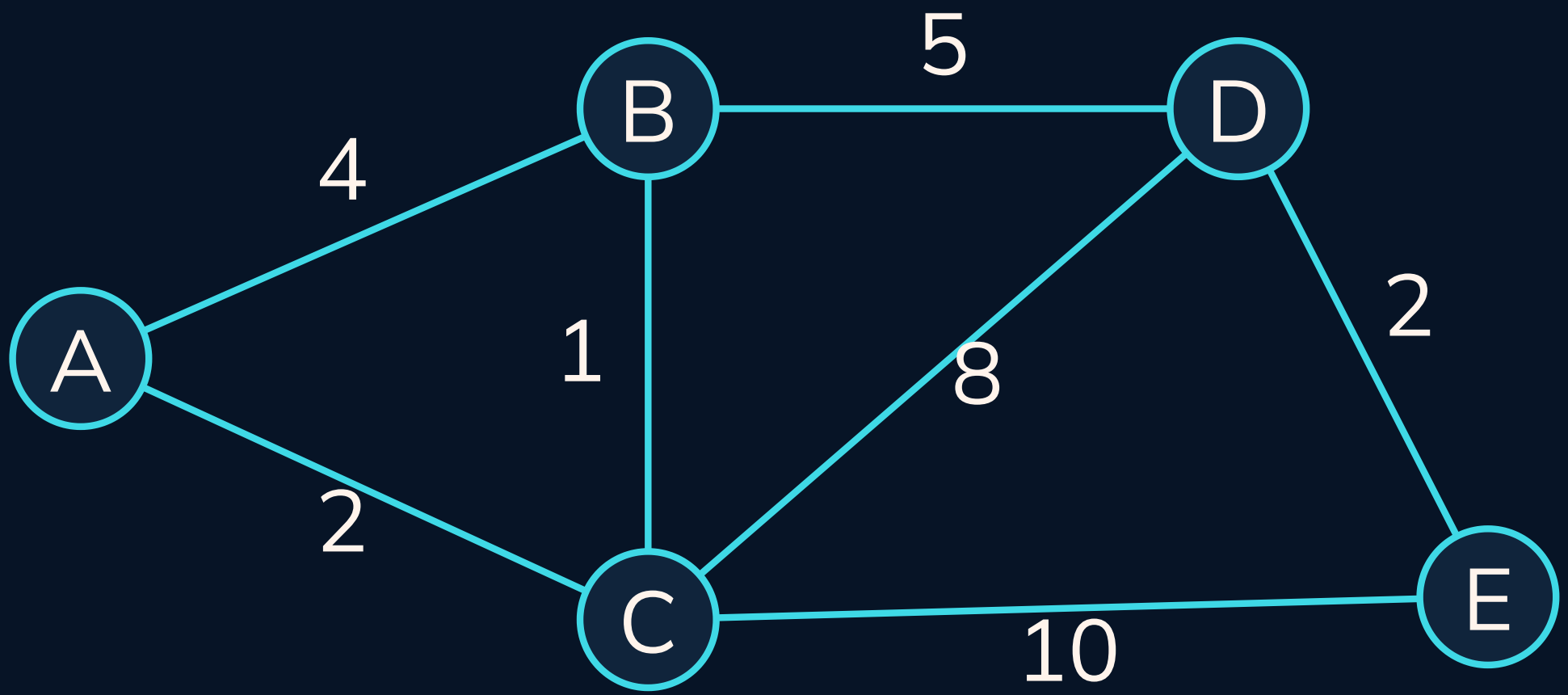
Avoid cycles by always reaching an unvisited vertex.



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# Our Graph

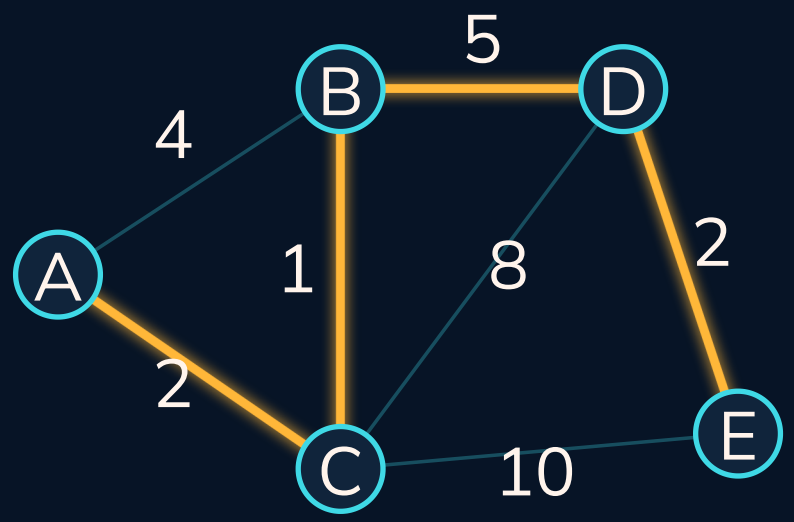
STARTING VERTEX = A




How can Prim’s algorithm connect all vertices with the minimum total cost?

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# Prim’s Algorithm: Step-by-Step



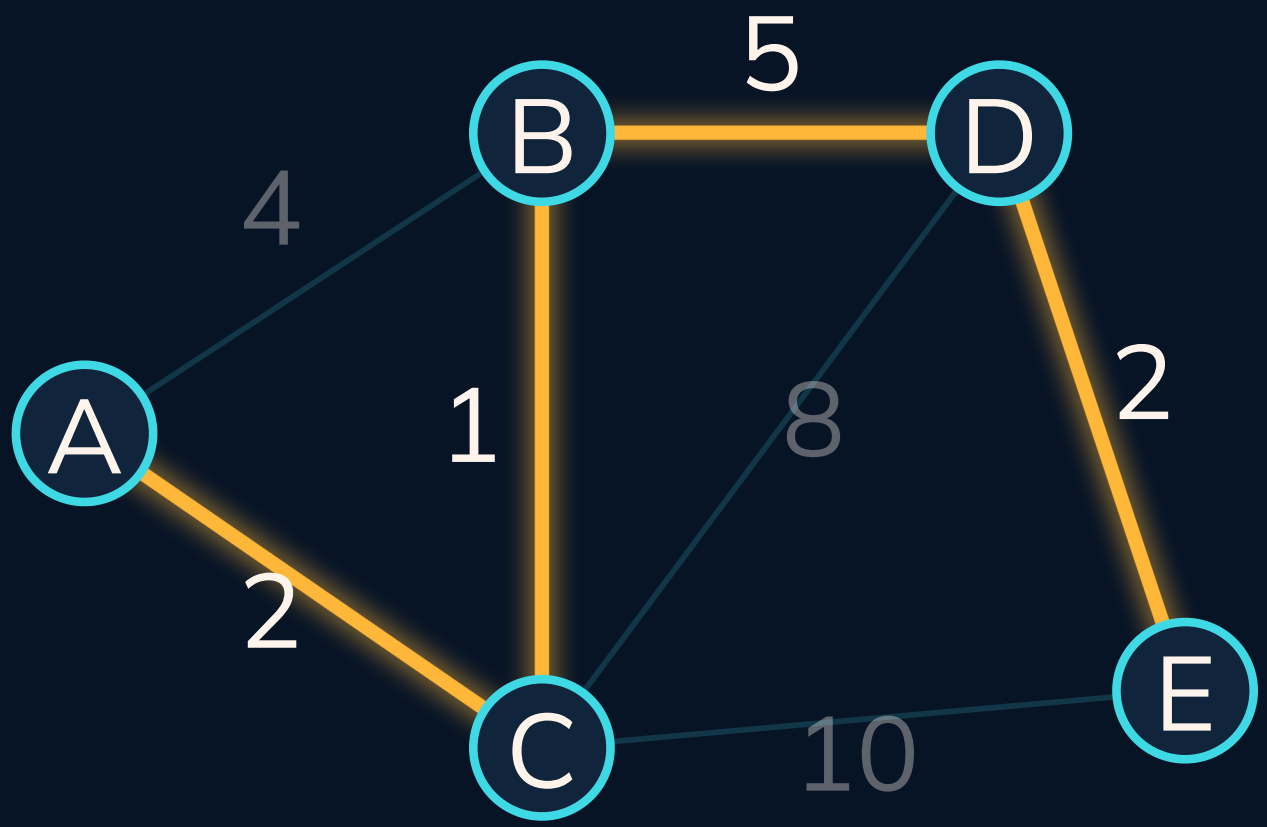

**A-C → C-B → B-D → D-E**

Start at A.
Add one new vertex at a time.

| Options | Choose | Visited |
|---|---|---|
| A-B = 4<br>A-C = 2 | A-C = 2 | A, C |
| From A, C:<br>C-B = 1 is minimum | C-B = 1 | A, C, B |
| B-D = 5<br>C-D = 8<br>C-E = 10 | B-D = 5 | A, C, B, D |
| Only E remains:<br>D-E = 2<br>C-E = 10 | D-E = 2 | All connected |

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# Minimum Spanning Tree Found!




Every vertex connected. No cycles.

MINIMUM SPANNING TREE COST

**10**

$$2 + 1 + 5 + 2 = 10$$

MST edges = 4

$$V - 1 = 5 - 1 = 4$$

Minimum possible total cost, without cycles.

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# Prim’s Algorithm · Pseudocode

```
PRIM(G, START)
```

1. Select a starting vertex
2. Mark it as visited
3. Find the minimum-weight edge connecting a visited vertex to an unvisited vertex
4. Add that edge to the MST
5. Mark the new vertex as visited
6. Repeat until all vertices are visited
7. Return the MST

KEY IDEA

Choose the cheapest edge connecting the current tree to a new vertex.



CHOOSE → ADD
VISIT → REPEAT

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# Complexity, Applications & Viva

## COMPLEXITY

Adjacency matrix: $O(V^2)$
Priority queue / min heap: $O(E \log V)$

Exact complexity depends on graph representation and data structure.



## APPLICATIONS

- Network design · efficient computer connections
- Road and transport · low-cost roads
- Electrical grids · minimum infrastructure cost
- Telecommunications · efficient networks
- Water and pipelines · reduced construction cost

## VIVA · HOW DO PRIM'S AND DIJKSTRA'S DIFFER?

| Prim's | Dijkstra's |
| --- | --- |
| Minimum Spanning Tree | Shortest paths from a source vertex |
| Minimum total tree cost | Minimum distance from a source |

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# THANK YOU

## Prim’s Minimum Spanning Tree Algorithm

“Connect everything.
Spend the minimum.”

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Created with the assistance of Gamma AI




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